GradeSense — Student Answer Script

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Question 1 — Science (5 marks)

A simple circuit is a complete conducting loop. The cell provides the potential difference that drives the current, the switch makes or breaks the loop, the resistor limits the current and the bulb converts electrical energy into light and heat. With the switch closed there is a continuous path and current flows; with it open the path is broken and the bulb goes out.

The ammeter is placed in the main loop, in series with the bulb and resistor, since the whole current has to pass through it to be measured. The voltmeter is connected in parallel, across the two ends of the bulb, because it compares the potential at either side of the bulb. Putting the voltmeter in the main loop would be wrong.

The arrow on my diagram shows conventional current leaving the positive terminal and returning to the negative terminal. Since $V = IR$ and the battery voltage is fixed, increasing R must reduce I. That is why adding resistance dims the bulb.

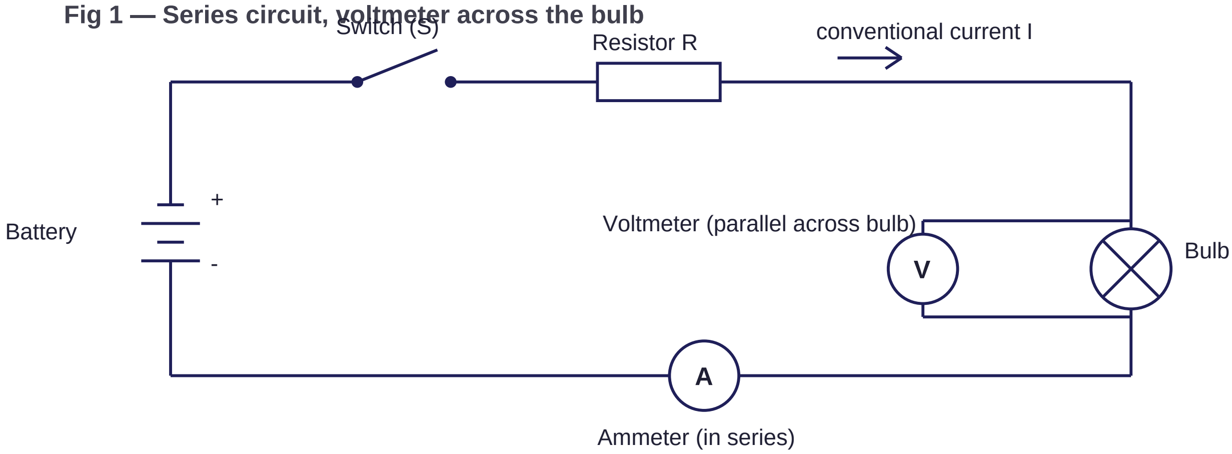

Fig 1 — Series circuit, voltmeter across the bulb

Question 2 — English (5 marks)

Access to information is not the same thing as learning, and I think that distinction is the heart of this statement. Technology has removed the barrier of finding out; it has not removed the work of understanding.

Where technology clearly helps is in repetition and choice. If one explanation does not work, a student can find another. When I was preparing for my mathematics test I used a graphing tool to see what changing a coefficient actually did to the curve, which no amount of re-reading the textbook had made clear to me.

The opposing case is real, though. A student who reaches for a search result at the first difficulty never sits with a problem long enough to learn from it, and struggling with a question is a large part of how understanding is built. Instant answers can quietly remove that struggle.

My conclusion is that technology raises the ceiling and lowers the floor at the same time. A student who already wants to understand gets far more out of it than before; a student who wants to be finished gets finished faster. The tool is the same, and the outcome depends on the intention behind it.

Question 3 — Economics (5 marks)

On my graph quantity is on the x-axis and price on the y-axis. Demand slopes down because people buy more at lower prices, and supply slopes up because producers offer more at higher prices. The curves cross at Rs 30 and 60 units. That is the equilibrium: at Rs 30 the quantity demanded and the quantity supplied are both 60, so the market clears exactly.

If the cost of production goes up, each unit is less profitable, so producers are willing to supply less at every price and the supply curve shifts to the left. With demand unchanged, the new intersection lies at a higher price and a lower quantity than before, which I have marked as E1.

Fig 2 — Demand and supply with shifted supply